# HW4 Q5

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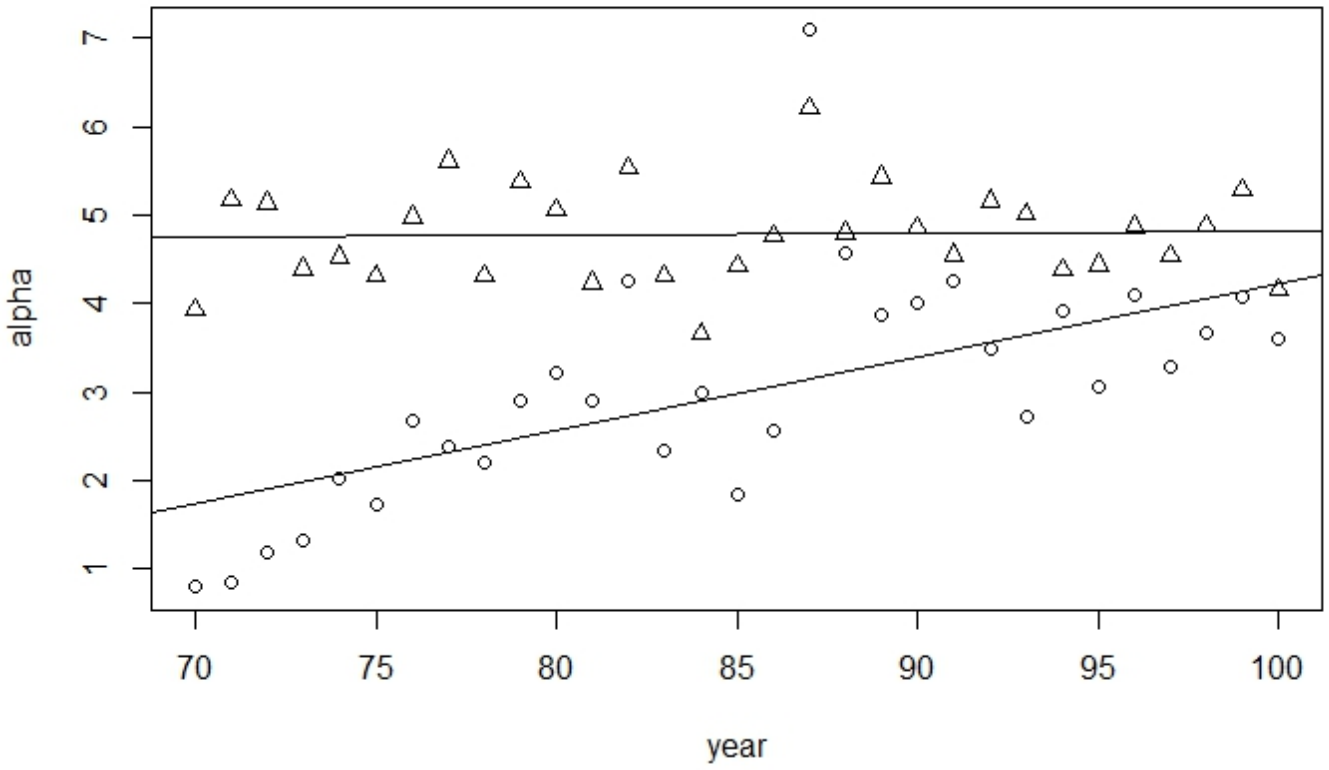
# Tail Behavior

- Apply Pareto Model:
  $F(x) = 1 - (\beta/x)^{\alpha}$
  positive trend indicates the field being less scientific
- Hill Estimator:
  $\hat{\alpha}_t = (m^{-1} \sum_{t=1}^{m} log(y_t^{(i)}/y_t^{(m+1)}))^{-1}$
- To correct bias with small smaple:
  $\tilde{\alpha}_t = \hat{\alpha}_t \hat{\alpha_N} / \bar{\alpha}_{nt}$

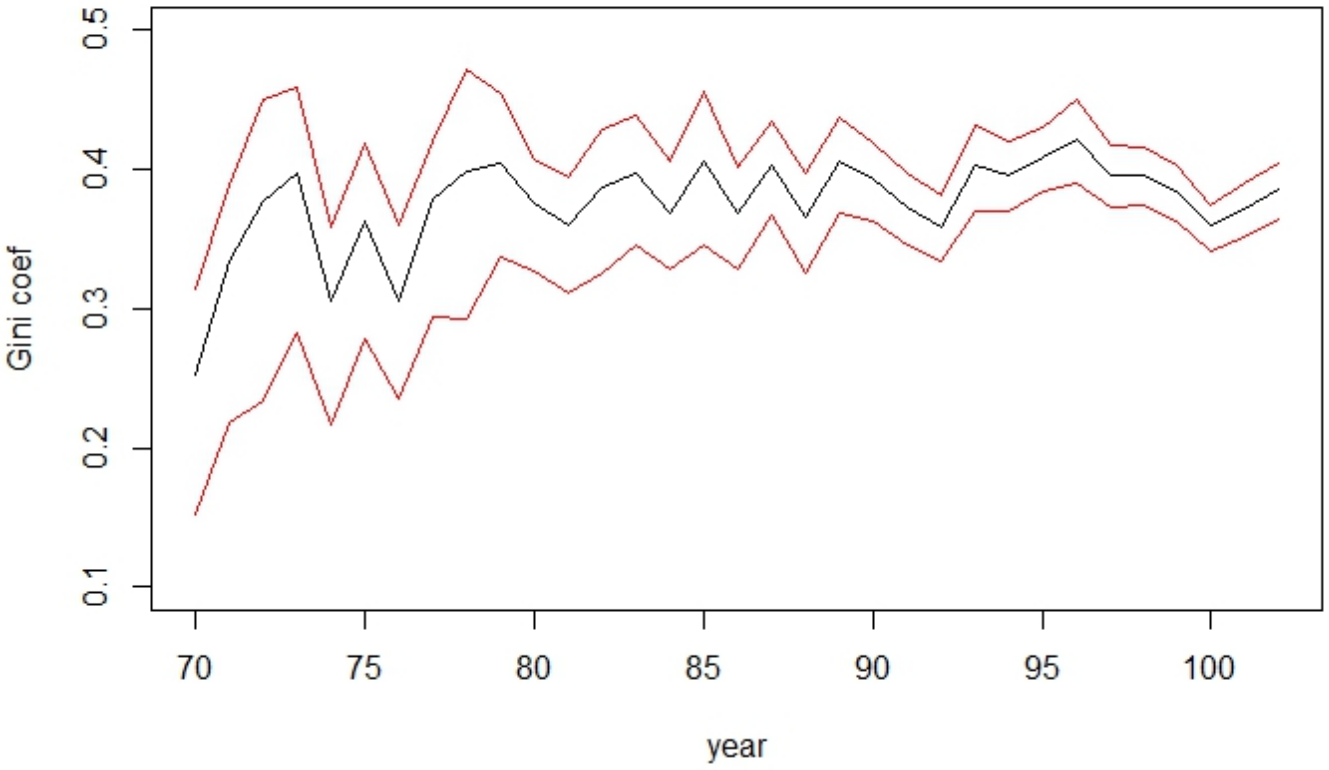
# Tail Behavior



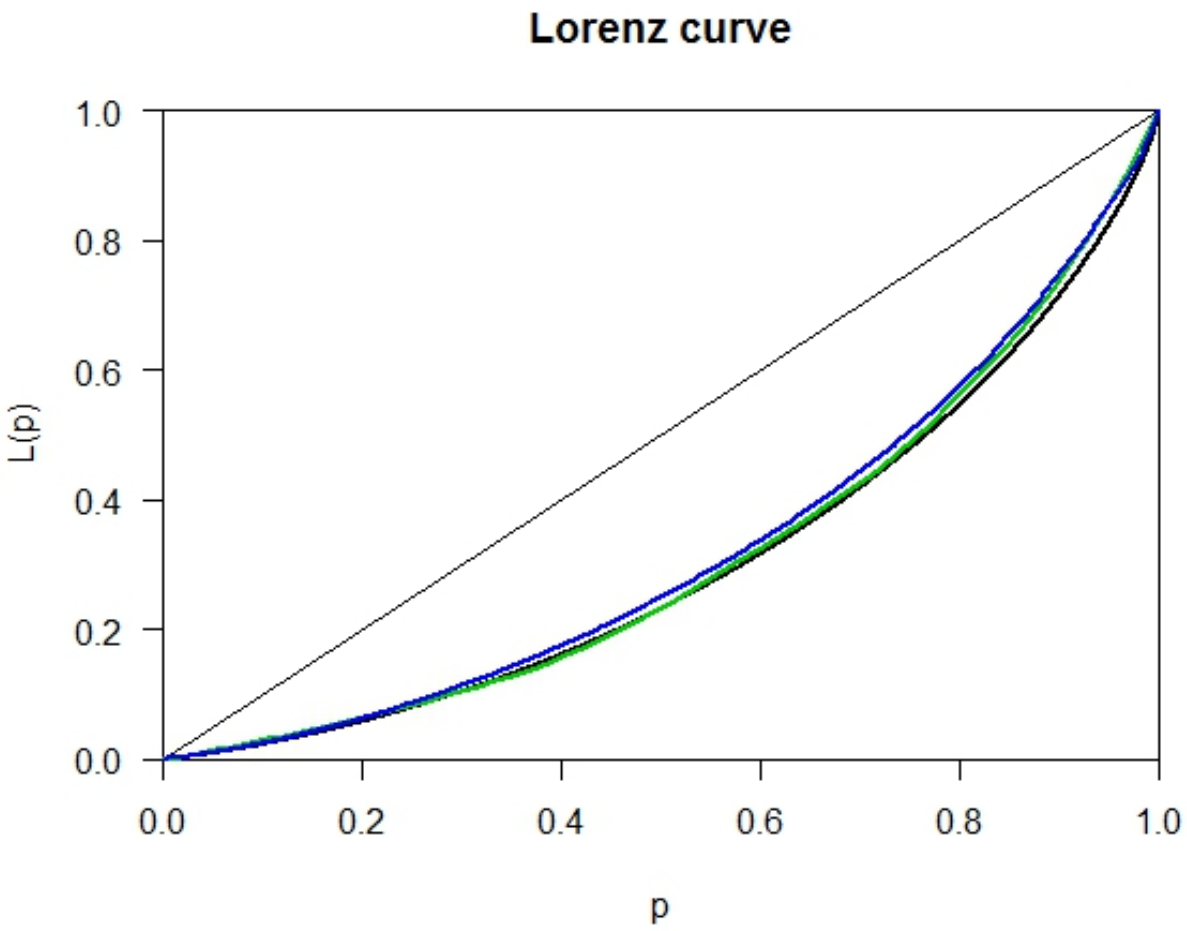
# Gini Coef

- An alternative way to look at this porblem
- Gini goes up means more inequality and fewer people are writing more paper. Then more scientific.

# Gini Coef



# Lorenz Curve




Lorenz curve at 1980, 2000 and for all years